ESE 123 Lab 4 Report: Function Generator and Oscilloscope

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Objective. Learn the 33210A function generator and the oscilloscope: how the internal 50 Ω affects the output, RMS of different wave shapes, and reading and triggering the scope.

1. Generator output vs. load (1 V_{rms} sine, 60 Hz)

The generator is an ideal source V_s in series with 50 Ω. It assumes a 50 Ω load, so when set to 1 V_{rms} it makes V_s = 2 V_{rms}:

$$V_{out} = V_s \cdot R_L / (50 + R_L)$$

Measurement	Display	DMM reading	% error
Q1: DMM only (step 4)	1.000 V_{rms}	2.005 V_{rms}	100.5 %
Q2: with 51 Ω resistor (step 6)	1.000 V_{rms}	0.996 V_{rms}	0.4 %

Measured load resistor (step 5): R_L = **50.13 Ω**.

Q3: $V_{out} = 2 \cdot 50.13 / (50 + 50.13) = 1.0013\ V_{rms}$
Compared with 0.996 V measured: −0.53 %

Q1: The error is about 100% because the DMM is about 10 MΩ, practically an open circuit. No current flows, so nothing drops across the internal 50 Ω and the DMM sees the full V_s = 2 V.

Q2 and Q3: With 50.13 Ω the load almost matches 50 Ω, so the output is about half of V_s, as the generator expects. The small remaining error comes from the resistor not being exactly 50 Ω and from instrument accuracy.

2. RMS of different wave shapes (1 V_{pp}, 51 Ω load)

$V_p = V_{pp} / 2 = 0.5$ V
Sine: $V_{rms} = V_p / \sqrt{2} = 0.354$ V Square (50%): $V_{rms} = V_p = 0.500$ V

Q4: Wave	Expected	DMM reading	% error
Sine, 1 V_{pp}	0.354 V_{rms}		
Square, 1 V_{pp} (−0.5 V to +0.5 V)	0.500 V_{rms}		

A square wave stays at its peak all the time, so its RMS is larger than a sine's.

3. Oscilloscope

Probe check. After Default Setup, both channels showed 10.0:1 attenuation. The 10:1 probe makes the scope look like 10 MΩ to the circuit.

Q5: 1 V_{pp} sine at 990 Hz on ch 2	Value
V_{pp} (counting boxes)	2 V_{pp}
Period (counting boxes)	≈ 1.05 ms

Q5: No, the scope shows about 2 V_{pp}, not the 1 V_{pp} setting. Like the DMM, the scope probe is about 10 MΩ, not 50 Ω, so almost all of V_s = 2 V_{pp} appears at the output.

Q6: Why one wave moves. Every sweep starts at the same point of the trigger signal, so it looks still. The other signal comes from a different instrument at a slightly different frequency, so its phase shifts a little each sweep and it drifts. When I switched the trigger source to channel 2, the sine stood still and the square wave drifted instead.